BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet

Cell Structure

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: ultrastructure, membranes and structure–function relationships.

A Retrieval

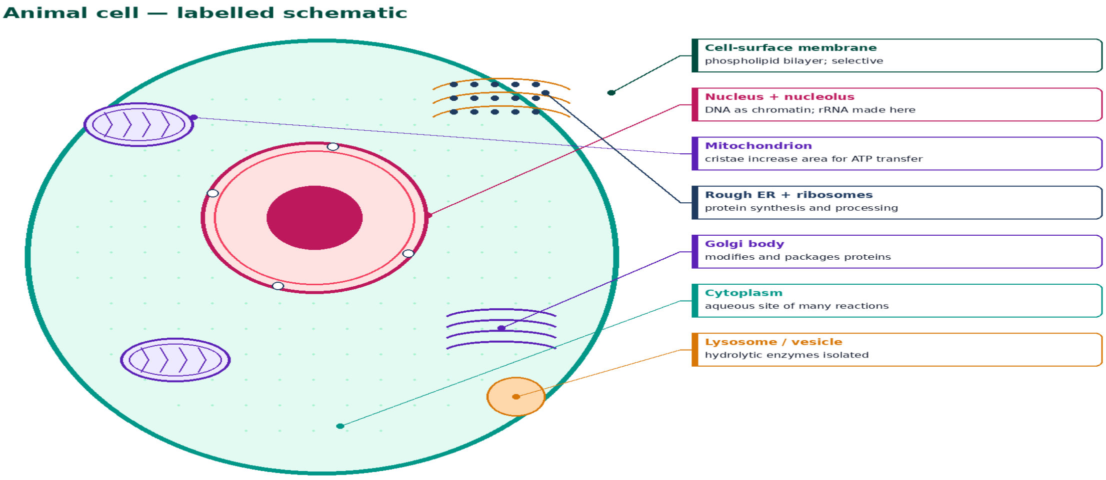
1. State the function of the nucleus, a mitochondrion and a ribosome.

2. Name three structures found in plant cells but not in typical animal cells.

3. What is meant by a selectively permeable membrane?

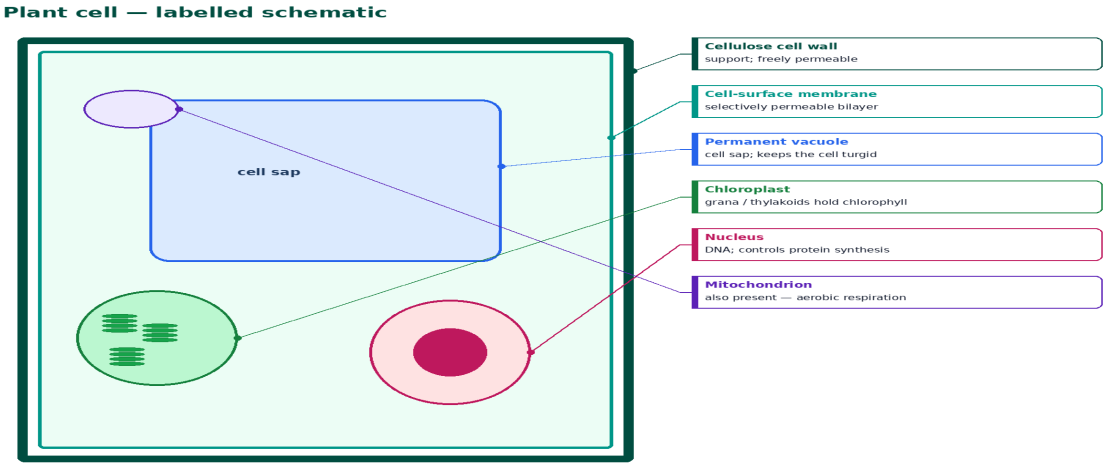
B Labelling

4. On the animal-cell schematic, name the structures indicated by the leader lines (or list six organelles and their functions).

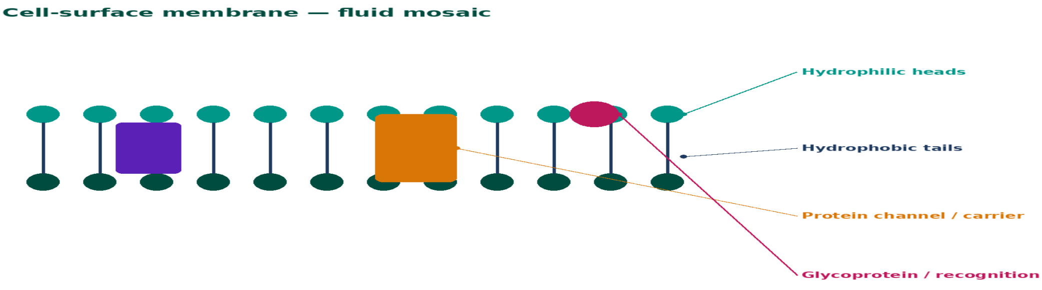


Original schematic — organelles are not to scale. Leader lines touch the structures.

5. Label the plant-cell extras: wall, membrane, vacuole, chloroplast and nucleus. State which layer is freely permeable.



The wall is freely permeable. Exchange is controlled by the membrane inside it.



Selectively permeable. Cholesterol (animal cells) regulates fluidity. Proteins form a mosaic.

6. Describe the structure of the cell-surface membrane. Use the words phospholipid, hydrophobic, protein and fluid mosaic.

7. Explain how cristae help mitochondria to transfer energy to ATP.

8. Why do muscle cells contain many mitochondria?

9. Root cells usually lack chloroplasts. Explain why this is not a problem for the plant.

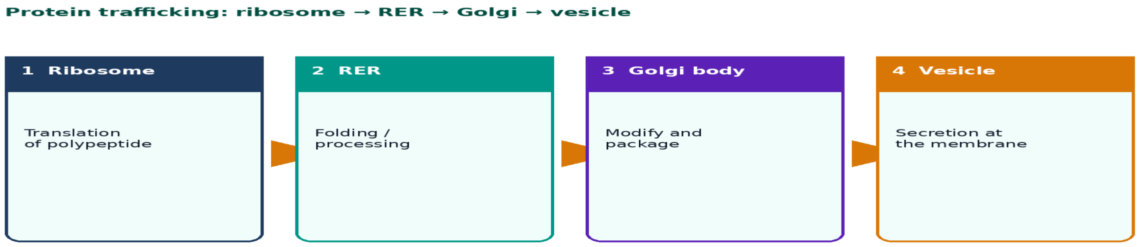
D Exam-style practice

10. Describe the functions of the nucleus and mitochondria in an animal cell. [4]

11. Explain two ways a palisade cell is adapted for photosynthesis. [4]

12. Distinguish between the cell wall and the cell membrane. [3]

E Challenge



13. Secretory cells have extensive RER and Golgi bodies. Explain this combination in terms of protein trafficking.